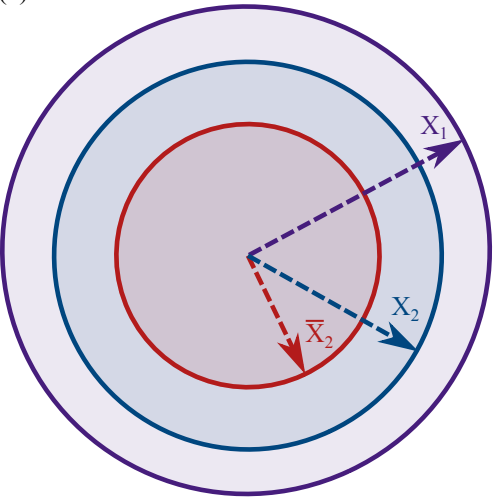
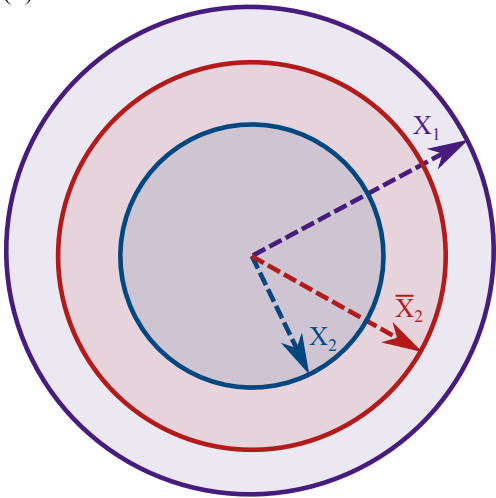
(a)



Case I: η = 1

(b)



Case II: η > 1